CCC-C-425a
September 26, 1961

SUPERSEDING
Fed. Spec. CCC-C-425
September 5, 1958

FEDERAL SPECIFICATION

CLOTH, COTTON, BIRDSEYE, AND GAUZE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements of two types of cotton cloth.

1.2 Classification.

1.2.1 *Types.* The cloth shall be of the following types as specified (see 6.2):

Type I.—Birdseye—single layer.

Type II.—Gauze—double layer.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

CCC-T-191—Textile Test Methods.

PPP-P-51—Packaging, Packing and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act, effective March 3, 1960.

(Copies may be obtained, free of charge, from the Federal Trade Commission, Washington 25, D. C.)

FSC 8305

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3. REQUIREMENTS

3.1 Material.

3.1.1 *Cotton yarn.* The cotton shall be cleaned and carded. The yarn shall be evenly spun to meet the requirements of this specification.

3.2 Color. The cloth shall be bleached white.

3.3 Physical requirements. The finished cloth shall conform to the requirements shown in table I or II as applicable.

3.3.1 *Weave.*

3.3.1.1 *Type I.* Type I cloth shall be a single-layer fabric in a birdseye weave.

3.3.1.2 *Type II.* Type II cloth shall be a double-layer gauze cloth woven in a plain, birdseye or interlocking-type weave. The combining, or interlocking, of the double layer in a single layer shall be throughout the length of the selvages and at regular intervals filling-wise to form pinking bars.

3.3.1.2.1 *Pinking bar.* A filling-wise bar (pinking bar) 1½ inches ± ½ inch wide shall be woven into the cloth at regular intervals so as to form panels of the double layer (2-ply cloth). The 1½-inch pinking bars shall be made by weaving the warps of the 2-ply cloth to form a single-layer cloth.

3.3.1.2.2 *Panels.* The panels of 2-ply cloth formed by the pinking bars shall measure 19 inches ± ½ inch in length from inside of pinking bar to inside of pinking bar.

3.3.1.2.3 *Folding stripes.* The contractor may, at his option, insert folding stripes. If folding stripes are used, one stripe consisting of four ends weaving as one in a group of 3 or 4 shall be positioned 13½ inches ± ½ inch from one selvage. Another stripe shall be positioned 8 inches ± ½ inch from the opposite selvage. The folding stripes shall become part of the single layer pinking bars at point of intersection.

3.4 Shrinkage. The sum of the warp and filling shrinkage for type I or type II cloth shall not exceed 20 percent when tested as specified in table IV.

3.5 Absorbency. Type I and type II cloths shall show a minimum absorption of 80 percent when tested as specified in table IV.

3.6 Finish. Type I and type II cloths shall be thoroughly cleaned, bleached, and shall contain not more than 2.0 percent total nonfibrous materials when tested as specified in table IV.

3.6.1 *Bleaching.* Type I and type II cloths shall be given a full chemical bleach. The use of optical bleaches and dyes is allowed subject to approval by the contracting officer.

3.7 Length. Unless otherwise specified, the cloth put up on rolls as specified in Federal Specification PPP–P–51 shall be furnished in one continuous piece of not less than 40 yards (see 6.2).

TABLE I.—*Physical requirements—type I, birdseye*

Weight, minimum (ounces/square yard)	Width (inches) (including selvages)	Yarns per inch, minimum		Breaking strength, minimum (pounds)	
		Warp	Filling	Warp	Filling
4.0	26½ minimum 27½ maximum	60	48	40	70

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TABLE II.—*Physical requirements—type II, gauze*

Weight[1], minimum (ounces/square yard)	Width (inches) (including selvages)	Yarns per inch, minimum[2]	Breaking strength[1], minimum (pounds)	
		Sum of Warp and Filling	Warp	Filling
3.35	39½ minimum 40½ maximum	74	45	25

[1] Utilizing two plys of panel.
[2] Determined on single ply of panel, but both single panels shall conform to this requirement.

3.7.1 *Pieces.* Pieces of type II cloth shall begin and end with a pinked edge located in the middle of the pinking bar.

3.8 Fiber identification. The cloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act effective March 3, 1960 (see 2.2).

3.9 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection for aceptance. Except when otherwise specified by civil government agencies, inspection shall be in accordance with Military Standard MIL–STD–105 except where otherwise indicated. Defects found during this examination shall be classified in accordance with 4.2.1, 4.2.2 and 4.2.3.

4.2.1 *Yard-by-yard examination.* The required yardage shall be examined on one side only, except that on type II the sides shall be alternated with each piece and visual defects classified as listed. All defects found shall be counted, except on type II only defects visible in the layer facing the examiner, shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only the more serious defects shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be one linear yard. The acceptable quality level (AQL) for this examination shall be 4.0 major defects and 15 total defects (major and minors combined) per 100 units (yards). The lot size shall be based upon level I of Military Standard MIL–STD–105. The lot size shall be expressed in units of one yard each. The number of pieces from which the sample is to be selected shall be in accordance with table III. An approximately equal number of yards shall be examined from each piece in the sample. Not more than one piece shall be taken from each container.

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Glossary Numbers Fed. Std. No. 4	Defects	Classification	
		Major	Minor
90, 91, 92	Hole, cut or tear in body of cloth	X	
78	Smash, any	X	
11, 33	Broken or missing yarns:		
	Three or more contiguous regardless of length ..	X	
	Two contiguous, or single 9 inches or more missing ..		X
9	Fine filling bar or light place, clearly visible[2]	X	
9, 12, 35	Open place, crack or hitchback: Opening at widest place four or more yarn diameters or sequence of openings resulting in weak spot	X	
13, 34	Floats or skips:		
	Multiple, ¾ inch or more in combined warp and filling directions	X	
	Multiple, less than ¾ inch in combined warp and filling directions, or single 2 inches or more in length ...		X
3, 5, 27	Slough-off, jerked in filling, slubs, slugs, or gouts—three or more times the thickness of normal yarns		X
6, 7, 28, 29	Kinks or knots, visible[2] accumulation		X
59, 61	Crease or wrinkle—hard, embedded		X
51	Selvage scalloped—deepest indentation more than ½ inch from straight edge of cloth		X
55, 56	Selvage cut, torn or broken	X	
74, 75, 76[1]	Spot, stain or streak ..		X
83	Trash — Motes, seed coat fragment or leaf — any visible[2] accumulation		X

[1] Except defects of spot, stain or streak shall be counted regardless of cause, i.e., not restricted to rust, oil or grease.

[2] At normal inspection distance (approximately 3 feet).

TABLE III.

Lot size in yards	Sample size in pieces	Maximum number of defects acceptable in sample
Up to 1,300[1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	1
22,001 to 110,000 ..	15	2
110,001 and over ..	25	2

[1] If a lot contains less than three pieces each piece in the lot shall be examined.

4.2.2 *Overall examination.* Each defect listed shall be counted not more than once in each piece examined. The unit of product for this examination shall be one piece. The sample size and acceptance number shall be as shown in table III.

Defects

Selvage curled, folded, rolled, or slack continuously or intermittently throughout the piece.
Overall uncleanliness, objectionable odor.
Width, type I less than 26½ or more than 27½ inches.
Width, type II less than 39½ or more than 40½ inches.
Uneven weaving throughout the piece.
Not bleached white[1].

Type II only:
Length of panels less than 18½ or more than 19½ inches.
Width of pinking bar less than 1 or more than 2 inches.
Folding stripes, if present, not in specified locations.

[1] A certificate of compliance shall be furnished with each shipment or lot, stating that the material has been fully bleached, using only bleaches and dyes approved by the contracting officer.

4.2.3 *Examination for length.*

4.2.3.1 *Individual pieces.* The pieces shall be examined for gross length. Any gross length found to be less than 40 yards, or any gross length found to be more than two yards below the gross length marked on the piece ticket or any type II not beginning or ending with a pinked pinking bar shall be considered as a defect with respect to length. The unit of product for this examination shall be one piece. The

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number of pieces selected as sample for this examination and the number of defects acceptable shall be in accordance with table III.

4.2.3.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual gross length of pieces in the sample is less than the total of the gross lengths marked on piece tickets. The pieces examined shall be those selected for examination of individual pieces.

4.3 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Examination	*Defect*
Marking (exterior and interior)	Omitted, incorrect, illegible, or improper size, location sequence, or method of application.
Materials	Any component missing.
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Weight	Gross weight exceeds requirements.

4.4 Testing of the end item. The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in table IV shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The unit of product shall be 3 continuous yards, full width, of the finished cloth. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

TABLE IV.—*Tests*

Characteristic	Requirement reference	Test method
Identification of cotton	3.1.1	1200
Weave	3.3.1	Visual[1]
Weight, ounce/square yard[2]	Tables I and II	5041
Yarns per inch[3]	Tables I and II	5050
Breaking strength[4]	Tables I and II	5100
Shrinkage in laundering[5]	3.4	5550
Absorbency	3.5	5502
Nonfibrous materials[6]	3.6	2611

[1] One determination per individual unit of product and results reported as "Pass or fail."

[2] For type II. Determination to be made on the two layers of body cloth and should not include folding stripes if present.

[3] For type II. Determination to be made on single layer of panel, specimens shall be sampled from both layers but both layers must meet requirement. Specimen should not include folding stripes if present.

[4] For type II. Determination to be made of double layer body cloth (not single ply). All tests must be clear of and must not include the pinking bars or folding stripes if present.

[5] The test shall be conducted in accordance with method 5550 except as follows:

Three specimens shall be prepared for test. The specimens of type I cloth shall be cut 26 inches in length, the full width of the cloth. The specimens of type II cloth shall be a complete diaper, double layer and shall be cut the width of the cloth through the middle of the respective pinking bars leaving a pinked edge. The samples shall be dried prior to pressing.

[6] The total nonfibrous material shall be determined by method 2611 utilizing the following formula:

$$\text{Nonfibrous materials content, percent} = \frac{O-S}{O} \times 100$$

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5. PREPARATION FOR DELIVERY

(Federal Standard No. 102 shall be referred to for definitions and application of the various levels of packaging and packing protection for supplies and equipment (for civil agencies only).

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A, B, and C.* The cloth put up on rolls shall be packaged in accordance with the applicable requirements for class 3 fabrics of Federal Specification PPP–P–51.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.2.1 *Levels A, B, and C.* Cloth of one type only shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of Federal Specification PPP–P–51.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in the fabrication of infants diapers.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type required (see 1.2).

(*c*) Length when other than specified (see 3.7).

(*d*) Option of using Military Standard MIL–STD–105 by civil government agencies (4.2).

(*e*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices and alternatives, as indicated in Federal Specification PPP–P–51, may be exercised in the preparation for delivery of cloth.

6.3 Transportation description. The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Cotton piece goods, finished, not otherwise indexed by name.
Carload minimum weight—Any quantity.

Motor:

Cotton piece goods, not otherwise indexed.
Motor volume minimum weight—Any quantity.

6.4 Level B packaging and packing should be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—M Q
Navy—S MC
Air Force—MAA

Copies of this specification may be purchased for 5 cents each.

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